

Hemalatha S

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PROFILE

Motivated engineering graduate with strong hands-on experience in Python, Java, full stack development, AI/ML, data science, and business intelligence tools. Interested in software development, machine learning, and data engineering roles, with the ability to build real-time applications and analytical solutions.

EDUCATION

B.E. Computer Science Engineering , <i>Jaya Sakthi Engineering College</i> CGPA: 7.9	2021 – 2025
HSC (Higher Secondary Certificate) , <i>Kamarajar Govt. Hr. Sec. School</i> 75%	2019 – 2021
SSLC , <i>Immaculate Hearts of Mary Hr. Sec. School</i> 70%	2017 – 2018

SKILLS & TOOLS

Programming Python, Java, SQL	Frameworks Django
Business Intelligence Power BI	Libraries NumPy, Pandas, Streamlit, OpenCV
Frontend HTML, CSS, JavaScript,react js	Tools VS Code, Eclipse, Anaconda, Google Colab

LANGUAGES

- English
- Tamil

PROJECTS

Autonomous CI/CD Code Repair Using Deep Reinforcement Learning , <i>Technologies: Python, PyTorch, Transformers, Deep Reinforcement Learning, GitHub Actions, CI/CD</i> Developed an autonomous software repair system to automatically detect and fix code errors during CI/CD pipeline execution. Used a Transformer encoder to understand failed code and error logs, and a DRL agent (DQN/A2C) to generate corrective patches. Integrated into CI/CD to reduce manual debugging and improve pipeline reliability. Demonstrated improvements in Fix Success Rate and MTTR using Defects] dataset.
Music Player Web Application , <i>Technologies: HTML, CSS, JavaScript, Figma, Canva</i> Built a responsive and interactive music player web application supporting smooth playback, playlists, and personalized suggestions. Designed modern UI/UX in Figma and implemented dynamic controls using JavaScript.
Fake News Detection on Multi-Modal Data (Unsupervised Learning) , <i>Technologies: Python, Streamlit, OpenCV, Tesseract OCR, DeepSeek, gTTS</i> Created a real-time fake news detection app for Instagram-like posts using image + text analysis. Implemented OCR, NLP, and clustering-based unsupervised learning to classify misinformation in health/finance content.

MERN Stack E-Commerce Web Application,

Tech Stack: MongoDB, Express.js, React.js, Node.js, JWT (Authentication), REST API.

"Developed a full-stack e-commerce application using the **MERN stack** (MongoDB, Express.js, React, Node.js). The project features a secure **User Authentication** system using JWT, a responsive front-end for product browsing, and a robust REST API backend. It manages real-time data storage in **MongoDB Atlas**, ensuring a seamless shopping experience with focus on performance and data security."

INTERNSHIPS

Data Science Internship

Worked on data preprocessing, feature engineering, visualization, ML model building, and deployment workflows.

Gradwin Software
Company

Java Internship

Gained practical experience in Java OOP, application logic, and problem-solving.

SoftGro Technologies

Python Internship

Worked on Python scripting, API handling, data processing, and backend development tasks.

V3 TechServ